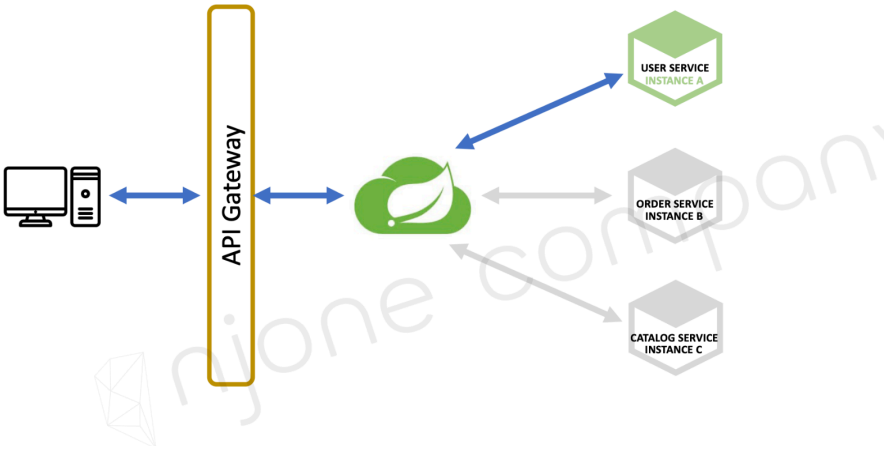
WEEK_5

Users MicroService(1)



프로젝트 생성 (user-service)

Dependencies

- Spring Boot DevTools
- Lombok
- Spring Web
- Eureka Discovery Client
- Spring Data JPA
- H2 Database

> https://github.com/bum0w0/spring-cloud-msa-study/commit/4760e6c41c0938006e9271ca01dda5b520b2e789

사용자 추가 (회원가입)

- POST /users
- UserController ↔ UserService ↔ UserRepository 흐름
- RequestUser(RequestDTO), UserDTO, UserEntity 데이터 교환

사용자 추가 구현 1 (with JPA)

- UserServiceImpl class에 createUser() 메소드 추가

```
@Override
public UserDto createUser(UserDto userDto) {
    userDto.setUserId(UUID.randomUUID().toString());

    ModelMapper mapper = new ModelMapper();
```

```
    mapper.getConfiguration().setMatchingStrategy(MatchingStrategies.STRICT);
    UserEntity userEntity = mapper.map(userDto, UserEntity.class);
    userEntity.setEncryptedPwd("encrypted_password");

    userRepository.save(userEntity);

    UserDto returnUserDto = mapper.map(userEntity, UserDto.class);

    return returnUserDto;
}
```

Http Status Code 변경

```
@PostMapping("/users")
    public ResponseEntity<ResponseUser> createUser(@RequestBody RequestUser user) {
        ModelMapper mapper = new ModelMapper();
        mapper.getConfiguration().setMatchingStrategy(MatchingStrategies.STRICT);

        UserDto userDto = mapper.map(user, UserDto.class);
        userService.createUser(userDto);

        ResponseUser responseUser = mapper.map(userDto, ResponseUser.class);

        return ResponseEntity.status(HttpStatus.CREATED).body(responseUser);
    }
```

Spring Security 연동

스프링 시큐리티 6.x 버전으로 넘어오면서 몇몇 기능은 권장되지 않음. 따라서, Step 2, 4, 6 은 실습에서 제외

- **Step 7: SecurityFilterChain을 반환하는 configure 빈 추가**

BCryptPasswordEncoder

- password 해싱을 위해 Bcrypt 알고리즘 사용
- 랜덤 Salt 부여 및 여러 번 Hash를 적용한 암호화 방식

UserServiceImpl.java

```
@Autowired
BCryptPasswordEncoder passwordEncoder;

@Override
public UserDto createUser(UserDto userDto) {
    userDto.setUserId(UUID.randomUUID().toString());
    userDto.setEncryptedPwd(passwordEncoder.encode(userDto.getPwd()));
```

UserServiceApplication.java

```
@Bean
public BCryptPasswordEncoder bCryptPasswordEncoder() {
    return new BCryptPasswordEncoder();
}
```

Utils.java (로그인 암호 생성)

```
public class Utils {
    public static void main(String[] args) {
        System.out.println(new BCryptPasswordEncoder().encode("password"));
    }
}
```

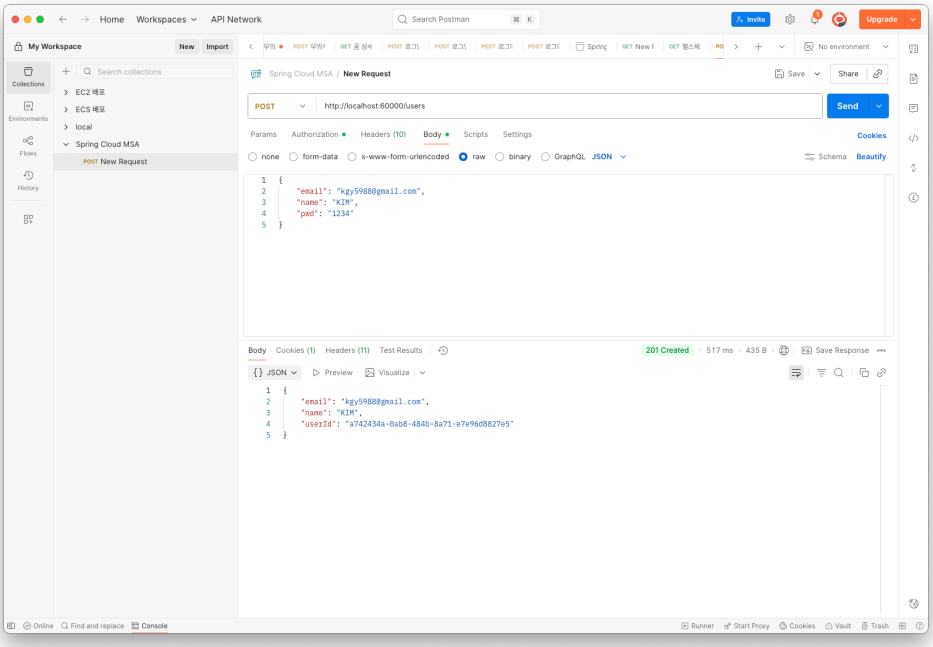
↓

```
$2a$10$4dUx4.6y8DnBpxhaFfj6lubIa5FeyMqQQM67XOogUNIDTdQ6nw4Qy
```

↓

```
security:
  user:
    name: user
    password: $2a$10$4dUx4.6y8DnBpxhaFfj6lubIa5FeyMqQQM67XOogUNIDTdQ6nw4Qy
```

Security 및 Bcrypt 적용 후 API 테스트


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
My Workspace
New
Import
Collections
Environments
Flows
History
EC2 배포
ECS 배포
local
Spring Cloud MSA
POST New Request
Spring Cloud MSA / New Request
Save
Share
POST
http://localhost:60000/users
Send
Params
Authorization
Headers (10)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"email": "kgy5988@gmail.com",
"name": "KIM",
"pwd": "1234"
}
Body
Cookies (1)
Headers (11)
Test Results
201 Created
517 ms
435 B
Save Response
JSON
Preview
Visualize
{
"email": "kgy5988@gmail.com",
"name": "KIM",
"userId": "a742434a-0ab8-484b-8a71-e7e96d8827e5"
}
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash